Intelligenza Artificiale

“Probabilistic reasoning over time (part 2)”

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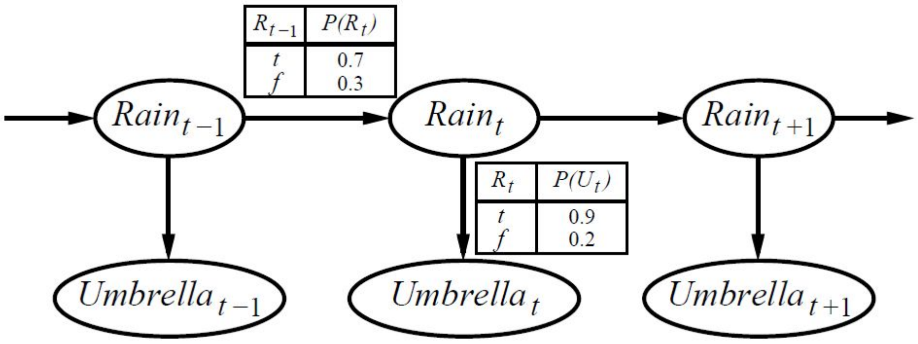
Lecture outline

- Hidden Markov Models
- Matrix representation
- Examples

Slides partly based on Russell & Norvig instructors material http://aima.cs.berkeley.edu/instructors.html

Hidden Markov Models

- **Hidden Markov model** (**HMM**): temporal probabilistic model in which the (hidden) state of the process is described by a single, discrete random variable
- No restriction on the evidence variables, there can be many
- E.g. umbrella world



Matrix representation of transition model

- For a state variable X_t of size S, the **transition model** $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t\text{-}1})$ becomes an S×S matrix $\mathbf{T}$ where:

$$\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$$

  and $\mathbf{T}_{ij}$ is the probability of a transition from state i to state j

- E.g., if X can take the values a, b, c, the model is a 3x3 matrix

$$\mathbf{T} = \begin{pmatrix} P(X_t{=}a \mid X_{t-1}{=}a) & P(X_t{=}b \mid X_{t-1}{=}a) & P(X_t{=}c \mid X_{t-1}{=}a) \\ P(X_t{=}a \mid X_{t-1}{=}b) & P(X_t{=}b \mid X_{t-1}{=}b) & P(X_t{=}c \mid X_{t-1}{=}b) \\ P(X_t{=}a \mid X_{t-1}{=}c) & P(X_t{=}b \mid X_{t-1}{=}c) & P(X_t{=}c \mid X_{t-1}{=}c) \end{pmatrix}$$

Matrix representation of sensor model

- Also the **sensor model** becomes an $S \times S$ <u>diagonal</u> matrix $\mathbf{O}_t$ where the elements of the diagonal are the probabilities of the observation e_t given any possible state X_t

$$P(e_t \mid X_t = i) \qquad \textbf{(emission probability)}$$

- For 2 possible observations, e.g. 0, 1, there are 2 diagonal matrices

$$\mathbf{O}_t^0 = \begin{pmatrix} P(e_t{=}0 \mid X_t{=}a) & 0 & 0 \\ 0 & P(e_t{=}0 \mid X_t{=}b) & 0 \\ 0 & 0 & P(e_t{=}0 \mid X_t{=}c) \end{pmatrix} \qquad \mathbf{O}_t^1 = \begin{pmatrix} P(e_t{=}1 \mid X_t{=}a) & 0 & 0 \\ 0 & P(e_t{=}1 \mid X_t{=}b) & 0 \\ 0 & 0 & P(e_t{=}1 \mid X_t{=}c) \end{pmatrix}$$

Filter with matrix representation

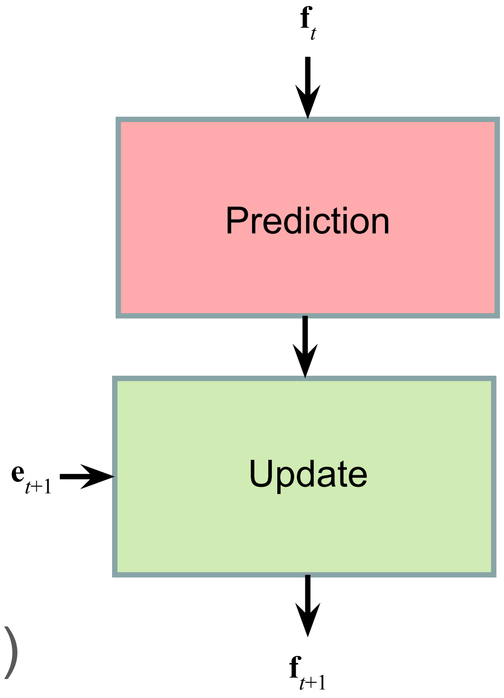
- Filtering

$$\begin{aligned}\mathbf{f}_t &= \mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_t) \\ &= \alpha_t\, \mathbf{P}(\mathbf{e}_t \mid \mathbf{X}_t)\, \Sigma_{\mathbf{x}_{t-1}} [\mathbf{P}(\mathbf{X}_t \mid \mathbf{x}_{t-1})\, f_{t-1}] \\ &= \alpha_t\, \mathbf{FORWARD}(\mathbf{f}_{t-1}, \mathbf{e}_t)\end{aligned}$$

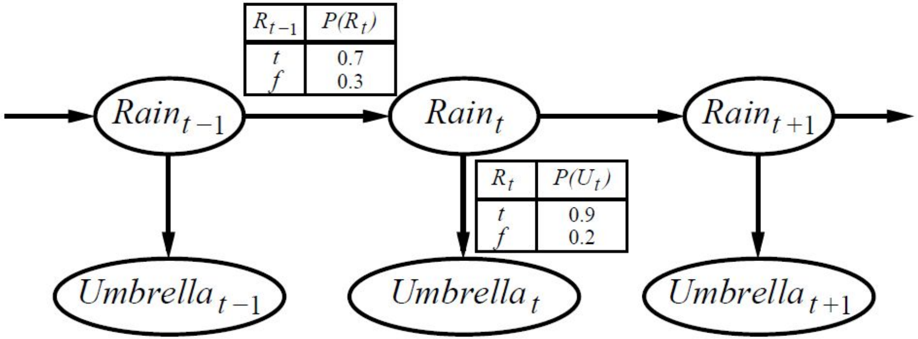
- The same equation can be rewritten as

$$\mathbf{f}_t = \alpha_t \boxed{\mathbf{O}_t} \boxed{\mathbf{T}^T \mathbf{f}_{t-1}}$$

($\mathbf{T}^T$ is the *transpose* of $\mathbf{T}$)



Umbrella world example



Rain

$$X_t = \begin{bmatrix} True \\ False \end{bmatrix}$$

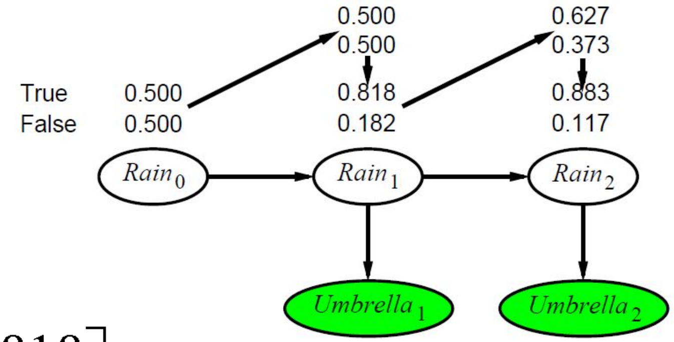
$$\mathbf{T} = \mathbf{P}(X_t \mid X_{t-1}) = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

Umbrella

a different $\mathbf{O}_t$ for each possible e_t

$$\mathbf{O}_t = \mathbf{P}(e_t = True \mid X_t) = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.2 \end{bmatrix}$$ and $$\mathbf{O}_t = \mathbf{P}(e_t = False \mid X_t) = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.8 \end{bmatrix}$$

Umbrella world example



$$\mathbf{f}_t = \alpha_t \mathbf{O}_t \mathbf{T}^T \mathbf{f}_{t-1}$$

in case $e_1 = True$

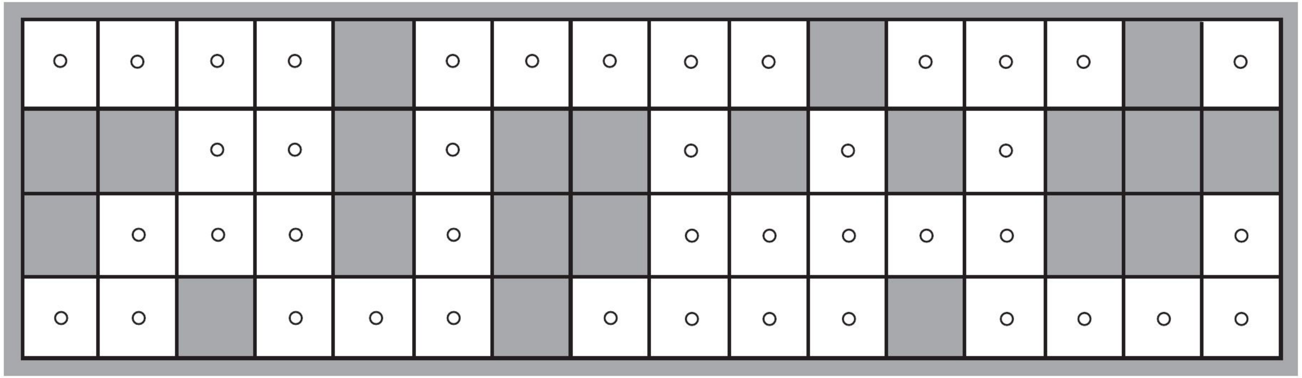
$$\mathbf{f}_1 = \alpha_1 \begin{bmatrix} 0.9 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \alpha_1 \begin{bmatrix} 0.45 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.818 \\ 0.182 \end{bmatrix}$$

in case $e_2 = True$

$$\mathbf{f}_2 = \alpha_2 \begin{bmatrix} 0.9 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.818 \\ 0.182 \end{bmatrix} = \alpha_2 \begin{bmatrix} 0.565 \\ 0.075 \end{bmatrix} = \begin{bmatrix} 0.883 \\ 0.117 \end{bmatrix}$$

Robot localization example

- PROBLEM: estimate the position of a robot with noisy sensors that moves randomly, with equal probability, to any adjacent empty square
- X_t is the robot's location on the grid, i.e any of the empty squares $\{1, \ldots, S\}$



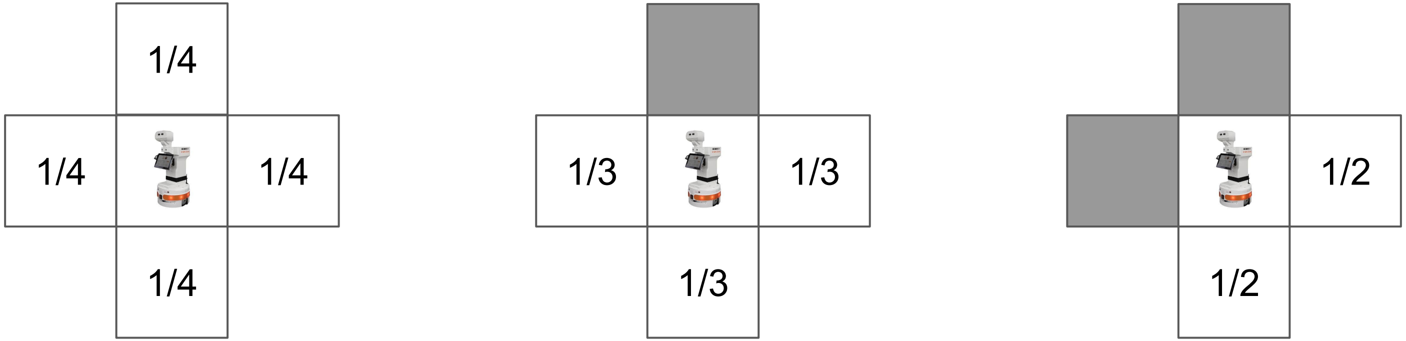
$S = 42$

Robot localization – transition model

- If $N(i)$ = number of empty squares adjacent to i, the transition probability is

$$P(X_{t+1} = j | X_t = i) = \mathbf{T}_{ij} = \begin{cases} 1/N(i) \text{ if } j \in \textsc{Neighbors}(i) \\ 0 \text{ otherwise.} \end{cases}$$

e.g.



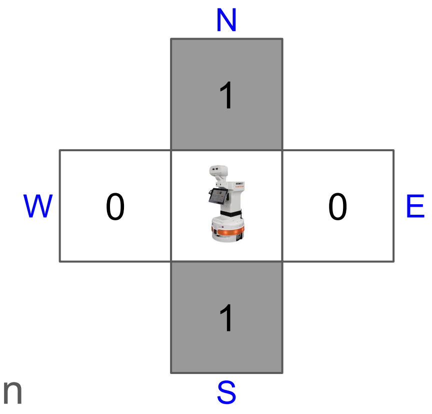
- Robot's initial position is unknown, so $P(X_0 = i) = 1/S$
- Since $S = 42$, the transition matrix $\mathbf{T}$ has $42 \times 42 = 1764$ entries

Robot localization – sensor model

- The sensor variable E_t has 16 possible values, each a 4-bit sequence giving the presence or absence of an obstacle in each of the compass directions N-E-S-W
  - e.g. 1010 means obstacle at North and South, free at East and West
- If the sensor's error rate ε (probability of wrong reading) is independent from direction, then
  - probability of getting all 4 bits wrong is ε^4
  - probability of getting them all right is $(1-\varepsilon)^4$
  - probability of getting 2 wrong and 2 right is $\varepsilon^2\,(1-\varepsilon)^2$
- In general, if d_{it} is the number of wrong bits in reading e_t at square i, then

$$P(E_t = e_t \mid X_t = i) = (\mathbf{O}_t)_{ii} = (1-\epsilon)^{4-d_{it}}\epsilon^{d_{it}}$$

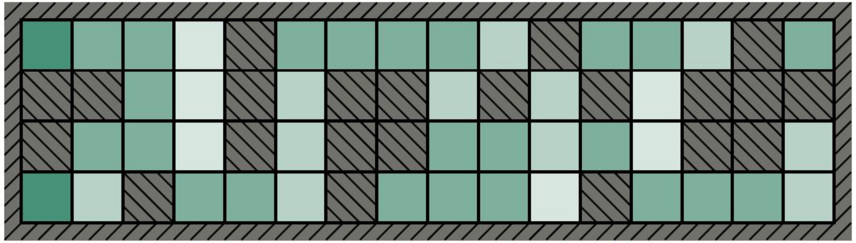
  - e.g. $P(E_t = \mathbf{1110} \mid X_t = \textbf{figure}) = (1-\varepsilon)^3\,\varepsilon$



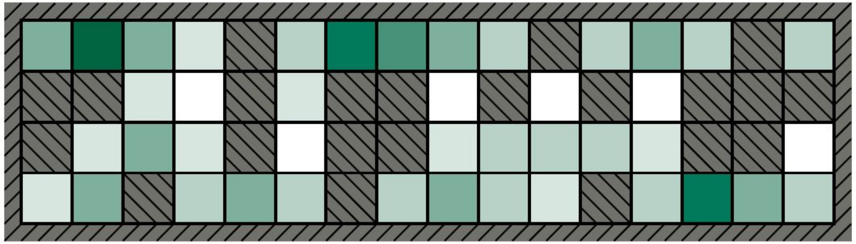
Robot localization example

- Given $\mathbf{T}$ and $\mathbf{O}_t$, we can use **Filtering** to compute the posterior distribution for each new sensor observation
  - e.g. $\mathbf{P}(X_1 \mid E_1 = \mathbf{1011})$
  - $\mathbf{P}(X_2 \mid E_1 = \mathbf{1011}, E_2 = \mathbf{1010})$
  - …

darker cell = higher probability



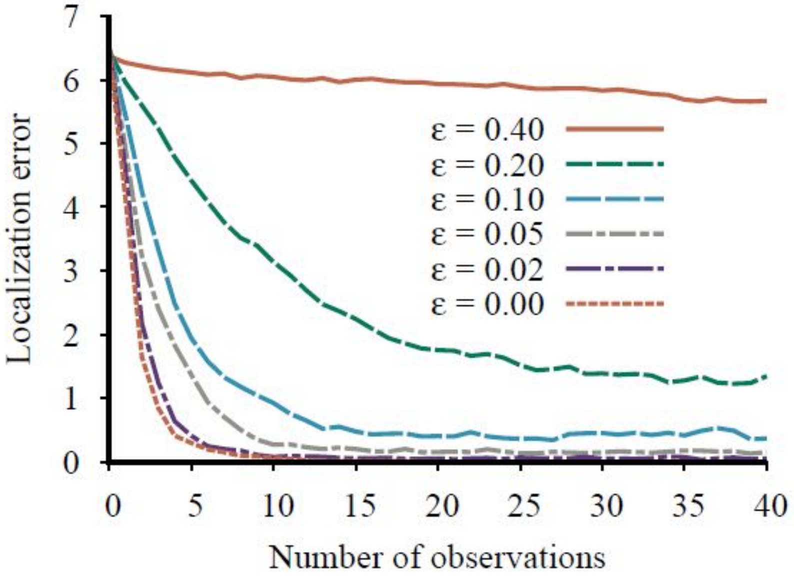
(a) Posterior distribution over robot location after $E_1 = 1011$



(b) Posterior distribution over robot location after $E_1 = 1011$, $E_2 = 1010$

Robot localization example

- Localization error for various values of the per-bit sensor error rate ε
- Even for $\varepsilon = 0.20$ (i.e. sensor reading wrong $1 - (1 - \varepsilon)^4 = 59\%$ of the time!) the robot is able to localise within 2 squares after 20 observations (green dashed lines in the graphs below)
- For $\varepsilon \leq 0.10$ the robot needs only a few observations to localise accurately
- For $\varepsilon = 0.40$, the localization error remains large even after many observations



More examples

Markov Model: grades example

- Consider a Bachelor degree, where at the end of each year students get a final grade: **A**, **B** or **C**
- The probability to have the same grade for two consecutive years is 0.6, and the probability to have a different grade is 0.2
- Each grade depends only on the one obtained in the previous year
- **QUESTION**: What is the probability of a student getting an **A** in year **3** if he/she got a **B** in year **1**?

Markov Model: grades example

- Let's call G_1, G_2, G_3 the three variables indicating the grades obtained in year 1, 2, 3, respectively

$$G_1 \rightarrow G_2 \rightarrow G_3$$

- We are asked to compute $P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_1 = \text{‘}\mathbf{B}\text{’})$

- Let's consider the general case $\mathbf{P}(G_3 \mid G_1 = \text{‘}\mathbf{B}\text{’})$

$$= \mathbf{P}(G_3, G_1 = \text{‘}\mathbf{B}\text{’}) \,/\, P(G_1 = \text{‘}\mathbf{B}\text{’}) = \alpha\, \mathbf{P}(G_3, G_1 = \text{‘}\mathbf{B}\text{’}) = \alpha \, \Sigma_{G2}\, \mathbf{P}(G_3, G_2, G_1 = \text{‘}\mathbf{B}\text{’})$$

$$= \cancel{\alpha}\, \Sigma_{G2} \mathbf{P}(G_3 \mid G_2)\, \mathbf{P}(G_2 \mid G_1 = \text{‘}\mathbf{B}\text{’})\, \cancel{P(G_1 = \text{‘}\mathbf{B}\text{’})} \qquad = \Sigma_{G2} \mathbf{P}(G_3 \mid G_2)\, \mathbf{P}(G_2 \mid G_1 = \text{‘}\mathbf{B}\text{’})$$

Markov Model: grades example

- Let's substitute some values

$$P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_1 = \text{‘}\mathbf{B}\text{’})$$

$$= \Sigma_{G2} P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_2)\, P(G_2 \mid G_1 = \text{‘}\mathbf{B}\text{’})$$

$$= P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_2 = \text{‘}\mathbf{A}\text{’})\, P(G_2 = \text{‘}\mathbf{A}\text{’} \mid G_1 = \text{‘}\mathbf{B}\text{’}) +$$

$$P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_2 = \text{‘}\mathbf{B}\text{’})\, P(G_2 = \text{‘}\mathbf{B}\text{’} \mid G_1 = \text{‘}\mathbf{B}\text{’}) +$$

$$P(G_3 = \text{‘}\mathbf{A}\text{’} \mid G_2 = \text{‘}\mathbf{C}\text{’})\, P(G_2 = \text{‘}\mathbf{C}\text{’} \mid G_1 = \text{‘}\mathbf{B}\text{’})]$$

$$= 0.6 * 0.2 + 0.2 * 0.6 + 0.2 * 0.2$$

$$= \mathbf{0.28}$$

HMM: baby example (sequence of observations)

- Consider a baby wearing a nappy that can be, with the same probability, “clean” or “dirty” (**hidden state**)
- The baby has a very limited vocabulary of four words: “mama”, “papa”, “poo-poo”, “pee-pee”
- At changing-time, if the nappy was “clean”, the probabilities of being “clean” or “dirty” at the next change are the same. If the nappy was “dirty”, the probability of being “clean” the next time is 0.8, and 0.2 to be “dirty” again

  (**transition probabilities**)

HMM: baby example (sequence of observations)

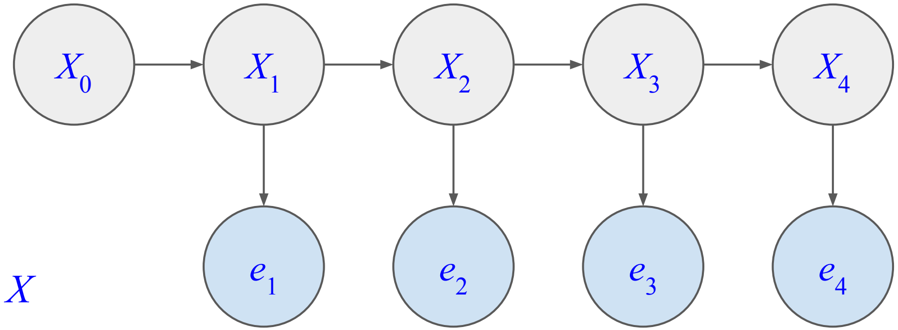
- Also, when the nappy is “clean”, the probabilities of the baby saying “mama” or “papa” are both 0.4, while the probabilities of saying “poo-poo” or “pee-pee” are both 0.1
- Vice-versa, when the nappy is “dirty”, the probabilities of the baby saying “mama” or “papa” are both 0.1, while the probabilities of saying “poo-poo” or “pee-pee” are both 0.4

  (**emission probabilities**)

- **QUESTION**: What is the probability of the baby saying the following words sequence: “mama” “mama” “pee-pee” “pee-pee” ?

  (given the usual Markov assumptions)

HMM: baby example



- Let's formulate the problem first
  - the nappy's content is the hidden state X
  - the spoken words are the emissions e
  - the transition probabilities $\mathbf{P}(X_t \mid X_{t-1})$ are

		X_t	
		clean	dirty
	clean	**0.5**	**0.5**
X_{t-1}	dirty	**0.8**	**0.2**

HMM: baby example

- The emission probabilities $\mathbf{P}(e_t \mid X_t)$ are

e_t

	mama	papa	poo-poo	pee-pee
clean	**0.4**	**0.4**	**0.1**	**0.1**
dirty	**0.1**	**0.1**	**0.4**	**0.4**

- We are asked to compute

$$P(e_1=\text{“mama”}, e_2=\text{“mama”}, e_3=\text{“pee-pee”}, e_4=\text{“pee-pee”})$$

HMM: baby example

- The probability of a sequence of words $\{e_1, e_2, e_3, e_4\}$ is equal to the sum of the probabilities of the sequence for each possible ending state x_4. That is

$$P(e_1, e_2, e_3, e_4) =$$

$$P(e_1, e_2, e_3, e_4, x_4 = \text{“clean”}) + P(e_1, e_2, e_3, e_4, x_4 = \text{“dirty”})$$

- More in general, calling $\mathbf{e}_\mathrm{t} = \{e_1, \ldots, e_t\}$

$$P(\mathbf{e}_\mathrm{t}) = \Sigma_{x_t} P(\mathbf{e}_t, x_t)$$

HMM: baby example

- Let's call $\mathbf{s}_t = \mathbf{P}(\mathbf{e}_t, X_t)$. Then

$$
\begin{aligned}
\mathbf{s}_t &= \Sigma_{x_{t-1}} \mathbf{P}(\mathbf{e}_t, x_{t-1}, X_t) \\
&= \Sigma_{x_{t-1}} \mathbf{P}(\mathbf{e}_{t-1}, e_t, x_{t-1}, X_t) \\
&= \Sigma_{x_{t-1}} \mathbf{P}(e_t \mid \cancel{\mathbf{e}_{t-1}, x_{t-1}}, X_t)\, \mathbf{P}(\mathbf{e}_{t-1}, x_{t-1}, X_t) \\
&= \mathbf{P}(e_t \mid X_t)\, \Sigma_{x_{t-1}} \mathbf{P}(X_t \mid \cancel{\mathbf{e}_{t-1}}, x_{t-1})\, P(\mathbf{e}_{t-1}, x_{t-1}) \\
&= \mathbf{P}(e_t \mid X_t)\, \Sigma_{x_{t-1}} \mathbf{P}(X_t \mid x_{t-1})\, s_{t-1}
\end{aligned}
$$

- … and we end up again with a recursive algorithm (like **filtering**, but in this case without normalization α)

HMM: baby example

- We can use the same matrix operations of filtering, but without normalization

$$\mathbf{s}_t = \mathbf{O}_t \mathbf{T}^T \mathbf{s}_{t-1}$$

$$\text{where} \quad \mathbf{T} = \mathbf{P}(X_t \mid X_{t-1}) = \begin{bmatrix} 0.5 & 0.5 \\ 0.8 & 0.2 \end{bmatrix}$$

$$\mathbf{O}_t^{mama} = \mathbf{O}_t^{papa} = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.1 \end{bmatrix} \quad \text{and} \quad \mathbf{O}_t^{pee-pee} = \mathbf{O}_t^{poo-poo} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.4 \end{bmatrix}$$

HMM: baby example

$$\text{case } e_1 = \text{"mama"}$$

$$\mathbf{s}_1 = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.260 \\ 0.035 \end{bmatrix}$$

$$\text{case } e_2 = \text{"mama"}$$

$$\mathbf{s}_2 = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0.260 \\ 0.035 \end{bmatrix} = \begin{bmatrix} 0.063 \\ 0.014 \end{bmatrix}$$

$$\text{case } e_3 = \text{"pee} - \text{pee"}$$

$$\mathbf{s}_3 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.4 \end{bmatrix} \begin{bmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0.063 \\ 0.014 \end{bmatrix} = \begin{bmatrix} 0.004 \\ 0.014 \end{bmatrix}$$

$$\text{case } e_4 = \text{"pee} - \text{pee"}$$

$$\mathbf{s}_4 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.4 \end{bmatrix} \begin{bmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 0.004 \\ 0.014 \end{bmatrix} = \begin{bmatrix} 0.001 \\ 0.002 \end{bmatrix}$$

HMM: baby example

- The probability of observing the sequence {"mama", "mama", "pee-pee", "pee-pee"} is therefore the sum of the elements of $\mathbf{s}_4$

$$P(e_1=\text{"mama"}, e_2=\text{"mama"}, e_3=\text{"pee-pee"}, e_4=\text{"pee-pee"})$$

$$= P(e_1, e_2, e_3, e_4, x_4 = \text{"clean"}) + P(e_1, e_2, e_3, e_4, x_4 = \text{"dirty"})$$

$$= 0.001 + 0.002$$

$$= \mathbf{0.003}$$

Conclusions

- Suggested reading
    - Chapter 14, sec. 14.3 of Russell & Norvig

- Next lecture

  Learning Bayesian Networks

- Questions?